

TUORUI PENG

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Education

Doctor of Philosophy in Statistics

Sep. 2023 – Present

Department of Statistics and Data Science, Northwestern University

Evanston, IL, USA

- Advisor: Prof. Matey Neykov

Master of Science in Statistics

Sep. 2023 – June. 2025

Department of Statistics and Data Science, Northwestern University

Evanston, IL, USA

Bachelor of Science in Mathematics and Physics

Sep. 2019 – June 2023

Department of Physics, Tsinghua University

Beijing, China

Bachelor of Science (minor) in Statistics

Mar. 2021 – June 2023

Department of Industrial Engineering, Tsinghua University

Beijing, China

Experiences

Graduate Researcher

Northwestern University

Project: Robust Inference under Shape Constraints with Heavy-Tailed Noise

Jan 2025 - Present

- Formulated a novel framework for robust mean estimation under star-shaped constraints, explicitly addressing non-Gaussian and heavy-tailed noise structures.
- Derived optimal minimax error bounds and established theoretical guarantees for estimators under non-convex and non-smooth geometric constraints.
- Presented findings in a comprehensive research paper ([arXiv:2604.05063](https://arxiv.org/abs/2604.05063)).

Undergraduate Researcher & RA

Tsinghua University

Project: Large-Scale Medical Record Dataset

Dec. 2021 – Jul. 2022

- Developed a scalable data pipeline to crawl, parse, and structure case-report articles from PubMed OA.
- Formed a large-scale public dataset of patient summaries and their links (160k patient summaries, 293k similarity annotations).
- Fine-tuned large language models (LLMs) to automate text mining and medical entity extraction from unstructured biomedical literature.
- Published the open-source dataset on [Nature Scientific Data](#). Demo see [Project website](#).

Undergraduate Dissertation

Tsinghua University

Project: Distribution Free Inference & NN Modeling for Gene Expression

Dec. 2022 – May 2023

- Applied Mixture Density Networks (MDN) to model complex, multi-modal conditional distributions for high-dimensional genetic data.
- Extended the Conformal Prediction framework to construct distribution-free conformal bands for conditional distribution functions.
- Programmed the entire pipeline in Python, achieving predictive coverage guarantees without relying on strong distributional assumptions.

Open Source Projects

- [High-Dimensional Statistics Digest](#): A rigorous, graduate-level reference text focusing on high-dimensional statistical inference and non-asymptotic analysis.
- [Mathematical & Statistical Foundations Summary Book](#): A technical repository summarizing advanced core methods, stochastic foundations, and computational techniques from undergraduate statistical curriculum.

Publications

Peng, T., Prasad, A., & Neykov, M. (2026). Robust mean estimation under star-shaped constraints with heavy-tailed noise. arXiv preprint arXiv:2604.05063.

Zhao, Z., Jin, Q., Chen, F., Peng, T., & Yu, S. (2023). A large-scale dataset of patient summaries for retrieval-based clinical decision support systems. Scientific data, 10(1), 909.

Honors & Rewards

Honorable Mention, Mathematical Contest in Modeling (2022)

Scholarship for Academic Advance (2020)

Skills

Programming: Python, R, C, L^AT_EX, Mathematica.

Languages: English (fluent), Chinese (native), Cantonese (fluent).